Bihar Engineering University, Patna
End Semester Examination - 2023

Course: B.Tech. | *Semester- II* | *Time: 03 Hours*
Code: 103202 | *Subject: Linear Algebra, Transform Calculus and Numerical Methods* | *Full Marks: 70*

Instructions:-
(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all*
(iv) *Question No. 1 is compulsory.*

Q.1 ***Answer any seven question of the following:*** **[2 x 7 = 14]**

(a) If A is a 3-rowed square matrix such that $|A| = 2$, then $|adj(adj(adj\ A^2))|$ is equal to
(i) 2^4 (ii) 2^8
(iii) 2^{16} (iv) None of the above

(b) Let A be a skew-symmetric matrix of order n, then
(i) $|A| = 0$, if n is even (ii) $|A| = 0$, if n is odd
(iii) $|A| = 0$, for all $n \in N$ (iv) $|A| \neq 0$, always

(c) If A is non-zero column matrix of the type $n \times 1$ and B is non-zero row matrix of the type $1 \times n$, then $\rho(AB)$ is
(i) 0 (ii) 1
(iii) n (iv) None of the above

(d) If $f(x) = 0$ is an algebraic equation, then Newton-Raphson method is given by
$x_{n+1} = x_n - \frac{f(x_n)}{?}$
(i) $f(x_{n-1})$ (ii) $f'(x_{n-1})$
(iii) $f'(x_n)$ (iv) $f''(x_n)$

(e) Gauss forward interpolation formula involves
(i) even differences above the central line and odd differences on the central line
(ii) even differences below the central line and odd differences on the central line
(iii) odd differences below the central line and even differences on the central line
(iv) odd differences above the central line and even differences on the central line

(f) What is/are the order(s) of the predictor and corrector in Milne's predictor-corrector method?
(i) 1 (ii) 2
(iii) 3 (iv) 4

(g) How many prior value(s) is/are required to predict the next value in Adams-Bashforth-Moulton method?
(i) 1 (ii) 2
(iii) 3 (iv) 4

(h) Laplace transform of unit step function
(i) $\frac{e^{-as}}{s}$ (ii) $\frac{e^{-as}}{s+1}$
(iii) $\frac{e^{as}}{s}$ (iv) $\frac{e^{as}}{s+1}$

(i) If $F(s)=L\{f(t)\}$, then $L\{f(\lambda t)\}=$
(i) $\lambda F(s)$ (ii) $\frac{1}{\lambda}F(\frac{1}{\lambda})$
(iii) $\frac{1}{\lambda}F(s)$ (iv) $F(\lambda s)$

(j) Trapezoidal rule can be applied if the number of equal sub-intervals of the interval of an interval is
(i) odd (ii) both
(iii) even (iv) None of the above

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Q.2 (a) Verify Cayley-Hamilton theorem for the matrix [8]

$$A=\begin{bmatrix}1 & 2 & 0\\ -1 & 1 & 2\\ 1 & 2 & 1\end{bmatrix}$$

Also obtain (i) A^{-1}, (ii) eigenvalues of A and A^2, and (iii) spectral radius of A.

(b) Diagonalize the matrix [6]

$$A=\begin{bmatrix}2 & 0 & 4\\ 0 & 6 & 0\\ 4 & 0 & 2\end{bmatrix}$$

By means of an orthogonal tranformation.

Q.3 (a) Derive Newton's forward interpolation formula. [7]

(b) Find the value of cos51°43' by Gauss's backward interpolation formula. Given that [7]

x	50°	51°	52°	53°	54°
$\cos x$	0·6428	0·6293	0·6157	0·6018	0·5878

Q.4 (a) Investigate, for what values of λ and μ do the system of equation [7]
$x+y+z=6$, $x+2y+3z=10$, $x+2y+\lambda z=\mu$ have
(i) no solution, (ii) unique solution and (iii) infinite solutions.

(b) Find the rank of the matrix [7]

$$\begin{bmatrix}3 & 2 & -1\\ 4 & 2 & 6\\ 7 & 4 & 5\end{bmatrix}$$

By reducing it to normal form.

Q.5 (a) Find [7]

$$\int_0^6 \frac{e^x}{1+x}dx$$

Approximately by using Simpson's 3/8 rule on integration.

(b) Evaluate [7]

$$\int_0^8 x\sec x\,dx$$

Using eight intervals by Trapezoidal rule.

Q.6 Using the Adams-Bashforth predictor-corrector equations, evaluate y(1.4), if y satisfies $\frac{dy}{dx}+\frac{y}{x}=\frac{1}{x^2}$ and y(1) = 1, y(1·1)=0·996, y(1·2) = 0.986, y(1·3) =0.972. [14]

Q.7 Solve the equation $u_{xx}=u_t$ subject to $u(x,0)=0$, $u(0,t)=0$ and $u(1,t)=t$, for two time steps by Crank-Nicolson method. [14]

Q.8 (a) Find the inverse Laplace transform of $\tan^{-1}(\frac{2}{s^2})$. [7]

(b) Evaluate $\int_0^\infty \{cost\cdot\delta(t-\frac{\pi}{4})\}\,dt$ by Laplace transform. [7]

Q.9 Write shorts notes on ***any two*** of the following: [7x2=14]

(a) Properties of Laplace transform

(b) Modified Euler's method

(c) Eigen values and Eigen vectors

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